CHALLENGE #2 · AI FOR AIR TRAFFIC MANAGEMENT

Three agents, one runway.

ATC, Pilot and Ground AI agents negotiate a disruption to a safe consensus in 3 turns, with a deterministic fallback and a tamper-evident log.

Disruptions are negotiated by phone, in silos

ATC

Plans landing slots around wake separation and a closed taxiway.

Pilot

Protects fuel reserve, safety and passenger connections.

Ground

Juggles gates, crew overtime and turnaround.

25% of today's uncoordinated plans, replayed on 300 disrupted 2015 flights, contain at least one hard-rule conflict.

Source: DOT 2015 flights.csv, organisers' constraints workbook; our evaluation of 300 rows

Separate agents, code-checked safety

ATC agent

Slots, runway, wake separation, throughput.

Pilot agent

Route, fuel margin, turbulence, connections.

Ground agent

Gate, crew, overtime, turnaround.

Constraint checker

Pure Python on every candidate and again before commit. LLMs never judge hard limits.

Arbiter

Urgency from world state, fuel first. Inflated claims are downweighted.

Fallback + log

Deterministic solver if no consensus. Hash-chained, append-only consensus log.

Each agent: its own LLM call, system prompt and private view; role-restricted levers

A bounded 3-turn protocol

TURN 1

Declare

Each agent sends one structured proposal. The checker tests candidate plans.

TURN 2

Vote

Accept or reject each feasible candidate with a reason. One counter-offer, own levers only.

TURN 3

Final vote

Only if needed. No new offers. Unanimous feasible plan wins.

The deterministic fallback decides and the log says why. The result is re-checked before commit.

Mean 2.01 turns across 300 scenarios; limit 3

Live case: DL1814 into Atlanta, in a storm

Situation. 2 of 4 runways closed, 22 arrivals/h, 10 other arrivals sequenced. Pilot is CRITICAL with a 5 min fuel margin.

Turn 1. ATC proposes 21:59 via WX_AVOID. The checker rejects the combined plan: fuel at landing 27 min is below the 30 min reserve.

Turn 2. The mediator repairs it. All three agents accept the repaired plan.

DECISION · CONSENSUS, TURN 2

Land **21:51** on 28R,
standard route
Gate **B2**, crew C1
0 min holding
0 missed connections

Real run with Claude agents: demo/1_storm_fuel_priority.jsonl; ATL 2015-06-09

Results on 300 disrupted flights

100%

consensus; 0% fallback; 2.01 mean turns

74 → 0

hard-rule conflicts vs today's process

523 → 60

crew overtime minutes

37 → 12

holding minutes (2,915 → 889 kg fuel)

2,604 → 1,807

connections missed

12 of 300

flights gave something up, mostly time traded for fuel margin

Offline agents, 300 rows; Claude agents on 20 rows: 20/20 consensus, 0 hard conflicts

WHY IT CAN BE TRUSTED

Agents negotiate. Code enforces safety. Every step is on the record.

5 fault injections

Fuel, ghost gate, wake, urgency and role faults are all caught.

Tamper-evident

Edit one old log entry and the hash-chain check fails.

Next

Real cited limits, multi-flight contention, controller UI pilot.

31 tests pass; replay mode runs the demo without an API key